Preliminary Report
Hurricane Isidore
24 September - 1 October 1996

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a. Synoptic History

Hurricane Isidore formed from a tropical wave that had a well-defined cyclonic circulation of clouds and was clearly marked at mid-levels in the Dakar sounding data when it crossed the west coast of Africa on 22 September 1996. Deep convection increased and satellite analysts provided the first Dvorak scale T-numbers on the 23rd, while the circulation passed to the south of the Cape Verde Islands. Thunderstorms became more concentrated, T-numbers from the NOAA Tropical Analysis and Forecast Branch (TAFB) increased to 2.0, and ship reports suggested the formation of a surface circulation by 1200 UTC on the 24th. The "best track" begins at that time (Fig. 1, Table 1), indicating the start of the tropical depression stage of Isidore.

The tropical cyclone was initially located to the south of a deep-layer anticyclone. It moved toward the west-northwest at 15-20 knots and intensified. An intense convective band wrapped around the center and the system became a tropical storm on the 25th. Further intensification ensued, an eye began to appear intermittently, and Isidore reached hurricane intensity on the following day.

The mature and dissipating stages of Isidore were influenced by a well-defined mid- through upper-level low that was quasi-stationary near 25°N 45-50°W through the 25th. The low then weakened and lifted northward to near 35°N, but was reinvigorated there by its interaction with a mid-latitude short-wave trough which passed by to the north on the 27-28th. The steering currents on the east side of the low gradually turned Isidore in a general northward direction. The forward speed slowed to about 10 knots during the turn on the 28th, but then increased to 20 knots on 1 October. During this period, Isidore reached its estimated maximum intensity of 100 knot winds. The eye disappeared on the 29th and upper-level westerly to southwesterly winds of around 60 knots contributed to a shearing and weakening of Isidore, down to a

tropical storm with an exposed low-level cloud center on the 29th, and then to a tropical depression on the 1st. Deep convection dispersed and Isidore transformed to extratropical status on the 2nd.

b. Meteorological Statistics

The "best track" (Table 1) was obtained from the data presented in Figs. 2 and 3. Those figures show Isidore's estimated central pressure and maximum one-minute wind speed, respectively, versus time. Position and intensity estimates from satellite pictures were provided by the NOAA Synoptic Analysis Branch (SAB) and TAFB, and by the Air Force Global Weather Central (AFGWC). They are the basis for showing Isidore's maximum wind speed at 100 knots and minimum pressure at 960 mb. Isidore passed through the eastern part of the NOAA drifting buoy network. Observations from those platforms helped define the western part of the cyclone's low-level wind field.

The ship *Magnific* reported southeast (140°) winds of 58 knots at 1200 UTC on the 30th of September, while located at 25.1°N 37.2°W, about 175 nm from the center of Isidore. The reliability of that measurement is in doubt because the estimated maximum surface wind near the center was 50 knots at that time. This was the only surface sustained wind report of 34 knots or higher to be possibly associated with Isidore.

c. Casualty and Damage Statistics

No reports of casualties or damages were received.

d. Forecast and Warning Critique

Table 2 lists track forecast error statistics. The official forecasts and many of the numerical models had track prediction errors that, on average, were at least 33% smaller than normal. The GFDL and GFDI performed best. CLIPER errors were very large at 72 hours.

NHC intensity forecasts were good, in general, but did not show fast enough weakening on the 27th and 28th.

Hurricane and tropical storm watches and warnings were neither issued nor necessary.

Table 1. Preliminary best track, Hurricane Isidore, 24 September - 1 October 1996.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
24/1200	8.6	23.3	1008	25	Trop. Depression
1800	9.2	25.2	1008	30	" "
25/0000	9.9	26.9	1006	30	" "
0600	10.3	28.5	1004	35	Tropical Storm
1200	10.7	30.0	1000	45	" "
1800	11.0	31.7	994	50	" "
26/0000	11.2	32.8	988	60	" "
0600	11.7	34.2	984	70	Hurricane
1200	12.4	35.8	980	70	" "
1800	12.8	37.1	977	75	" "
27/0000	13.1	38.6	974	80	" "
0600	13.7	39.8	972	85	" "
1200	14.1	41.1	968	90	" "
1800	14.9	42.2	963	95	" "
28/0000	15.8	42.9	960	100	" "
0600	16.7	43.1	961	100	" "
1200	17.8	43.2	965	95	" "
1800	19.0	43.2	968	90	" "
29/0000	20.0	42.9	973	85	" "
0600	20.8	42.2	979	75	" "
1200	21.8	41.3	982	70	" "
1800	22.6	40.9	986	65	" "
30/0000	23.8	40.4	989	60	Tropical Storm
0600	25.3	40.1	992	60	" "
1200	26.7	40.0	995	50	" "
1800	28.2	40.1	997	45	" "

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1/0000	29.9	40.3	1000	35	Tropical Storm
0600	31.6	40.4	1002	35	" "
1200	33.6	40.4	1005	30	Trop. Depression
1800	35.8	40.3	1008	30	" "
2/0000	38.1	40.2	1010	25	Extratropical
0600	40.3	39.6	1013	20	" "
1200	43.0	36.0	1015	20	" "
28/0000	15.8	42.9	960	100	Min. Pressure

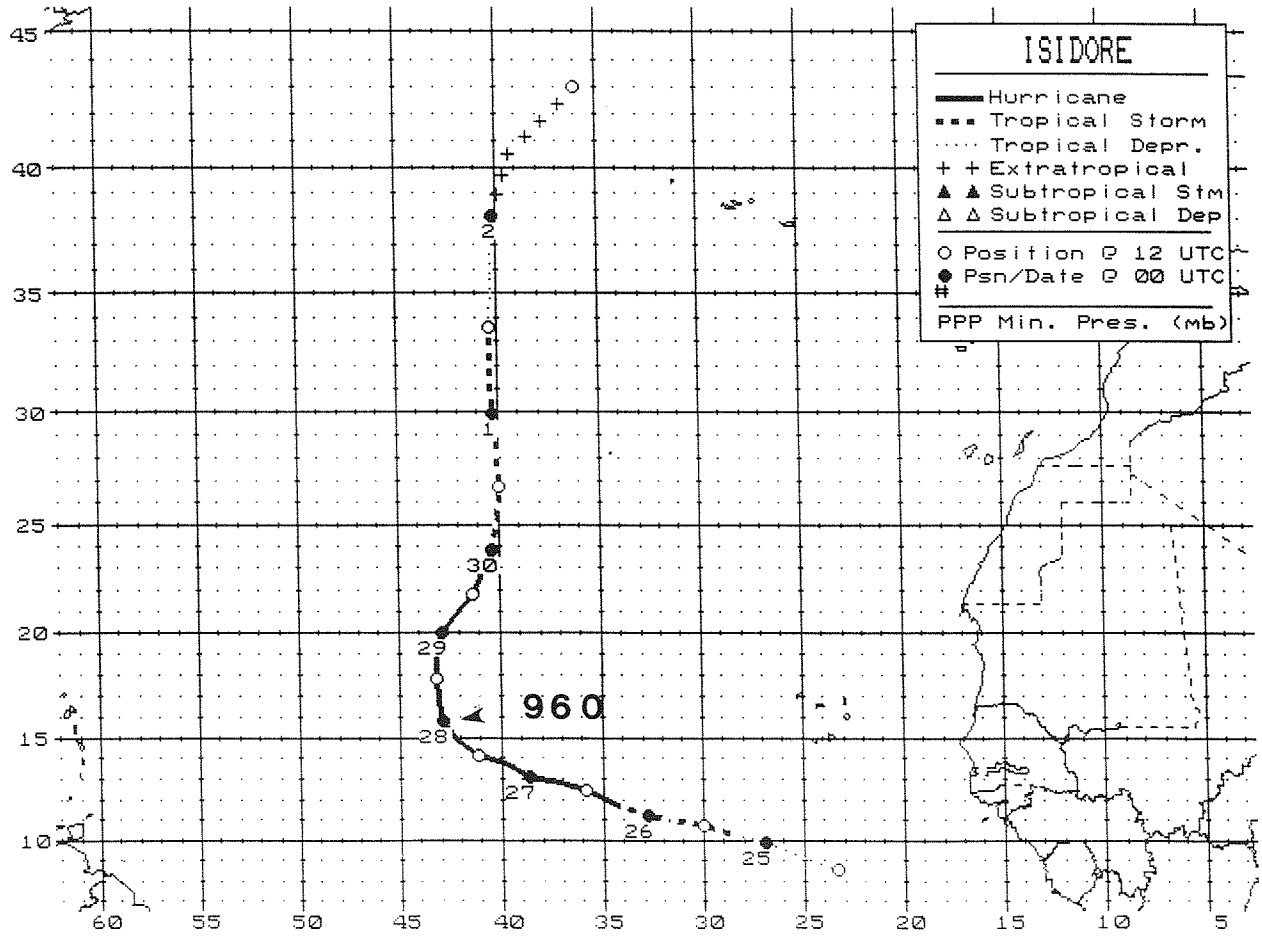

Figure 1. Best track positions for Hurricane Isidore.

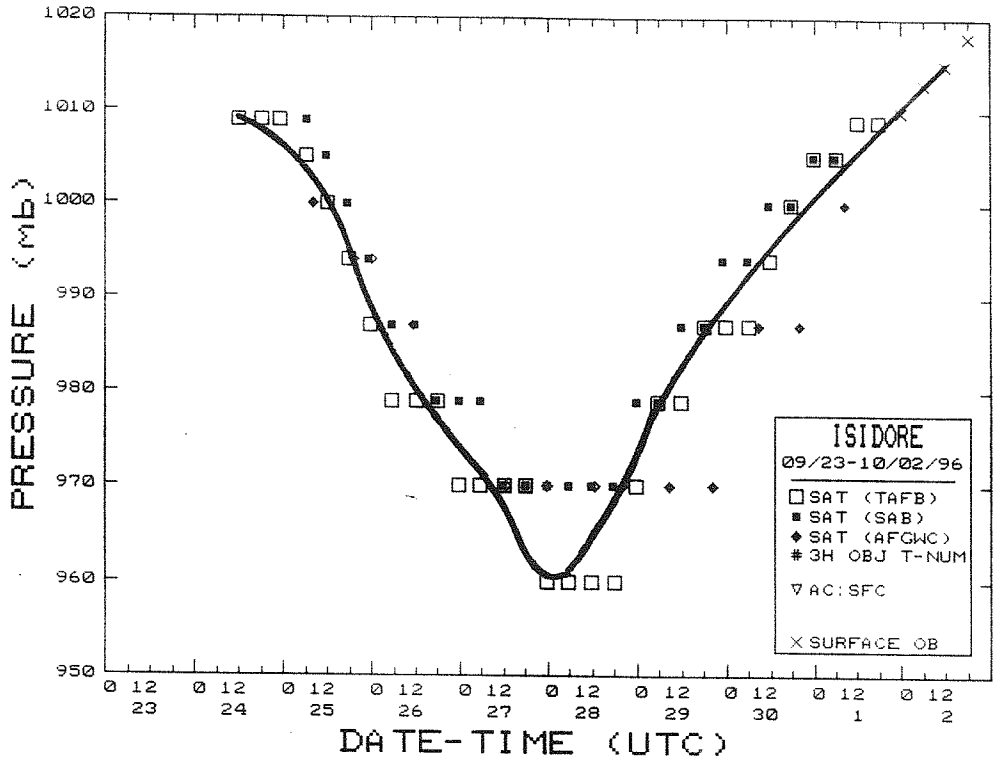

Figure 2. Best track central pressure curve for Hurricane Isidore, September-October 1996.

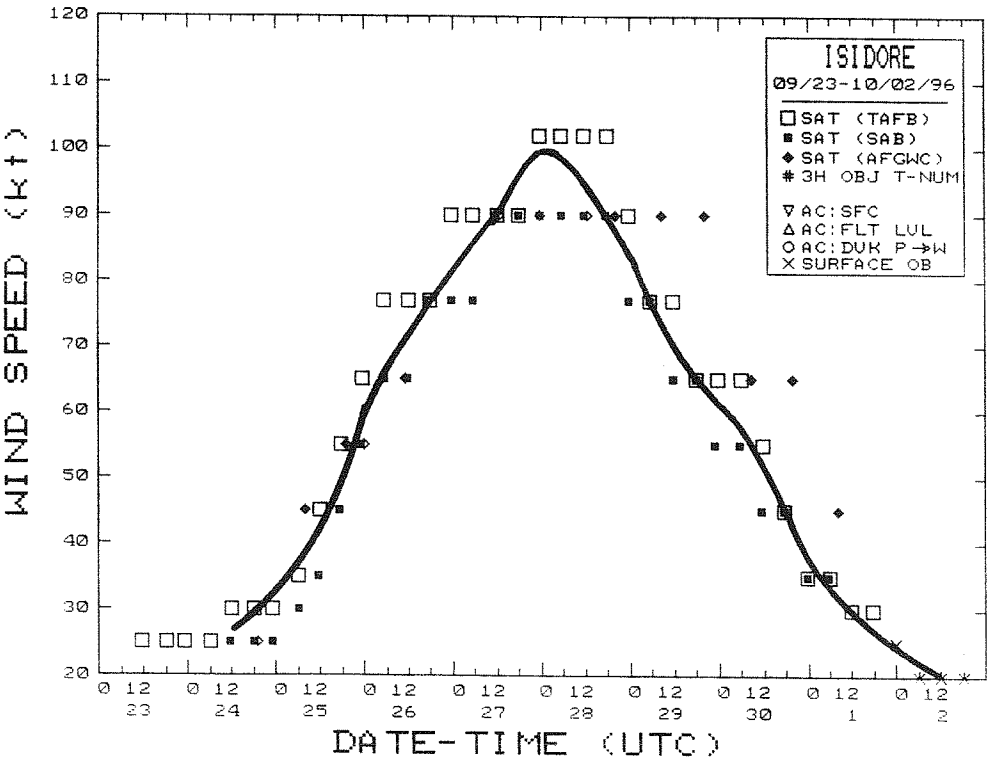

Figure 3. Best track maximum one-minute wind speed curve for Hurricane Isidore, September-October 1996.

Table 2

Preliminary forecast evaluation of Hurricane Isidore
Heterogeneous sample

(Errors in nautical miles for tropical storm and hurricane stages with number of forecasts in parenthesis)

Technique	Period (hours) 12	24	36	48	72
CLIP	56 (23)	128 (21)	210(19)	300 (17)	607 (13)
GFDI	37 (23)	62 (21)	81 (19)	102 (17)	133 (13)
GFDL*	34 (11)	55 (10)	81 (9)	98 (8)	149 (6)
VBAR	38 (22)	70 (21)	99 (19)	112 (17)	185 (13)
LBAR	37 (22)	64 (20)	85 (18)	107 (17)	216 (13)
AVNI	61 (21)	110 (19)	176 (17)	235 (16)	216 (12)
BAMD	35 (23)	64 (21)	96 (19)	126 (17)	183 (13)
BAMM	46 (23)	81 (21)	123 (19)	165 (17)	229 (13)
BAMS	60 (23)	110 (21)	160 (19)	202 (17)	278 (13)
A90E	48 (23)	96 (21)	151 (19)	185 (17)	274 (13)
NGPI	55 (19)	96 (18)	146 (16)	137 (13)	222 (9)
UKMI	55 (12)	113 (10)	184 (9)	227 (9)	272 (8)
NHC OFFICIAL	44 (23)	74 (21)	100 (19)	125 (17)	202 (13)
NHC OFFICIAL (1986-1995 10-year average)	49 (1670)	93 (1484)	136 (1314)	181 (1155)	273 (882)

* GFDL output not available until after forecast issued. VBAR output sometimes not available until after forecast issued.